

MBA ZG583 Final Student Packet

Management of AI Products Capstone

Course: MBA ZG583 - Management of AI Products

Assessment Type: Experiential Learning Assignment / Capstone

Recommended Weightage: 30%

Team Size: 3 to 5 students

Audience: Working professionals pursuing the MBA

How to Use This Packet

This is the **single student-facing document** for the capstone.

It includes:

- the full assignment brief,
- the expected deliverables,
- practical templates,
- evaluation guidance,
- recommended cases and demos,
- and a sample strong submission outline in the appendix.

Use this as your main working reference from topic selection to final submission.

Part A. Assignment Brief

1. Purpose of This Assignment

This assignment is designed to help you think and work like an **AI Product Manager**, not just like a student writing a report.

Your team must:

- identify a meaningful business or workflow problem,
- decide whether AI is genuinely the right lever,
- define the product strategy,
- create strong product-management artifacts,
- build a prototype or MVP,

- and defend your recommendation like a team seeking approval for pilot or launch.

This assignment also expects you to use AI tools in a disciplined way while doing PM work.

In other words, you must show:

1. **AI in the product**
 2. **AI in the product-management process**
-

2. Learning Goals

This assignment integrates the full course, especially:

- identifying and prioritizing AI product opportunities
- translating business problems into AI product requirements
- using PM frameworks such as JTBD, MVP, prioritization, and roadmap logic
- working across data, model, UX, API, and business layers
- evaluating AI systems using both technical and business metrics
- understanding lifecycle, scaling, cost, ethics, and governance
- using AI tools responsibly while producing PM artifacts

By the end of a strong project, your team should be able to demonstrate:

- an AI use case that is tied to a real business or workflow problem
 - a reasoned product recommendation, not just an interesting demo
 - a prototype that makes the core user or operator experience tangible
 - evidence that you can connect PM judgment, technical realism, and business value
 - disciplined thinking about human oversight, trust, and launch readiness
-

3. Project Options

You may choose one of the following:

Option A: Bring Your Own Product

Choose:

- a new AI-powered SaaS product

- a generative AI application
- an AI assistant or copilot
- an AI feature for an existing product
- an AI-enabled enterprise workflow

Option B: Choose from the Instructor Product Catalog

Select one approved product idea from the course materials.

Option C: Improve a Real Workflow

This is strongly recommended for working professionals.

Possible areas:

- support operations
- recommendation or personalization
- sales or retention
- HR or skills mapping
- fraud or risk screening
- forecasting or demand planning
- procurement or operations
- compliance or governance workflow

Expected weekly commitment

This capstone is designed for busy working professionals, but it is still substantial.

Plan for roughly:

- 2 to 3 hours per week per person during topic selection and framing
- 3 to 5 hours per week per person during artifact creation and prototype work
- additional time near final submission for integration, polish, and rehearsal

The strongest teams do not leave prototyping, metrics, or risk analysis to the final week.

4. What Makes a Strong Topic

Strong topics usually have:

- a specific user or business problem
- repeated decisions or repeated friction

- measurable value
- realistic access to data, workflow knowledge, or proxy evidence
- a clear MVP path

Weak topics usually sound like:

- “We want to build a chatbot”
- “We want to use AI because it is popular”
- “We want to automate everything”

Rule of thumb

Start with the **workflow problem**, not the technology.

AI use-case validation questions

Before committing to a topic, ask:

- is there a repeated decision, judgment, prediction, or content task here?
 - what is hard, slow, expensive, or inconsistent in the current workflow?
 - why is AI better than a simple rule, dashboard, search tool, or process fix?
 - what happens when the model is wrong?
 - where does human review remain necessary?
 - what is the smallest credible MVP that still creates value?
-

5. What You Must Submit

Your team must submit **one integrated assignment package**.

Final submission must include:

1. Executive Summary
 2. Product Strategy and Opportunity Framing
 3. PM Artifact Pack
 4. UX / Workflow / Trust Design
 5. Data, Model, and Evaluation Strategy
 6. Business, Economics, and Scaling View
 7. Ethics, Governance, and Risk Note
 8. Functional Prototype / MVP
 9. AI Collaboration Log
 10. Individual Contribution Disclosure
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6. Recommended Final Submission Structure

Use the following structure:

```
TeamName_Assignment/  
  01_Executive_Summary.pdf  
  02_Final_Report.pdf  
  03_Presentation_Deck.pdf  
  04_Prototype_Readme_or_Link.md  
  05_AI_Collaboration_Log.md  
  06_Individual_Contribution_Statement.pdf  
Appendix/  
  PRD/  
  Wireframes/  
  Metrics/  
  Roadmap/  
  Risk_Note/  
  Screenshots_or_Video/
```

If your prototype is online, include:

- repository link
- live demo link
- screenshots or short video
- setup instructions if needed

7. Deliverable-by-Deliverable Guide

7.1 Executive Summary

Purpose

Give a leadership-quality overview of the project.

Maximum size

- 2 pages or
- 4 slides

Must include

- business problem
- target user/customer
- product idea
- why AI is the right lever
- your recommendation:
 - explore
 - prototype
 - pilot
 - scale
 - reject

Template

```
## Executive Summary
```

```
### Problem
```

```
[What problem are you solving?]
```

```
### Target User / Customer
```

```
[Who has this problem?]
```

```
### Product Concept
```

```
[What is the proposed AI product / feature / workflow?]
```

```
### Why AI?
```

```
[Why is AI better than a traditional non-AI alternative here?]
```

```
### Recommendation
```

```
[What should happen next and why?]
```

7.2 Product Strategy and Opportunity Framing

Purpose

Show that the project is worth doing and worth doing now.

Must include

- product vision

- user problem / JTBD
- opportunity framing
- build vs buy vs partner/API decision
- first wedge / first use case

Recommended frameworks

- JTBD
- AI Opportunity Matrix
- RICE
- Build vs Buy vs API
- Platform vs Feature

Template

Product Strategy

Product Vision

[1-2 sentence vision]

JTBD

When I am [situation], I want to [motivation], so that I can [outcome].

Opportunity Framing

- Why this problem matters
- Why current alternatives are weak
- Why AI may help

Opportunity Matrix

Idea	Business Value	Feasibility	Priority
---	---	---	---

Build vs Buy vs Partner/API

- Recommended path:
- Why:
- Main trade-offs:

First Wedge

[What narrow use case will you start with first?]

7.3 PM Artifact Pack

Purpose

Show that you can create real PM artifacts, not just a pitch.

Minimum required artifacts

- problem statement
- product vision
- PRD or PRD-lite
- feature prioritization logic
- MVP definition
- roadmap
- success metrics

Important expectation

You must show **how these artifacts change when the product is AI-powered.**

Template

```
## PM Artifact Pack

### Problem Statement
[What problem, for whom, in what context?]

### Product Vision
[What future state are you creating?]

### PRD Summary
- User problem
- Core workflow
- AI role
- Risks
- Fallback logic

### Feature Prioritization
| Feature | User Value | Effort | Confidence | Priority |
|---|---|---|---|---|
| | | | | |

### MVP Definition
[What is the minimum version that proves something important?]

### Roadmap
- Phase 1
- Phase 2
- Phase 3

### Success Metrics
- Product metric
- Business metric
- Trust / guardrail metric
```

7.4 UX, Workflow, and Trust Design

Purpose

Show that you understand AI UX is probabilistic, not deterministic.

Must include

- wireframes or flow
- key user journey
- uncertainty handling
- trust / explainability features
- fallback / escalation / HITL where needed

Template

```
## UX and Workflow Design

### Main Workflow
[Describe the user journey]

### Wireframes / Screens
[Attach screenshots or links]

### Trust and Explainability
- What does the user need to know?
- What must be visible?
- What should be hidden?

### Uncertainty Handling
- High confidence behavior:
- Low confidence behavior:
- Failure or fallback behavior:

### Human-in-the-Loop
[Where does human review or override enter?]
```

7.5 Data, Model, and Evaluation Strategy

Purpose

Show that your AI logic is operationally credible.

Must include

- problem translated into an AI/ML task
- data sources

- data assumptions
- labeling logic if relevant
- baseline alternative
- model / API / rules choice
- evaluation framework
- launch or go/no-go thresholds

Template

```
## Data, Model, and Evaluation Strategy

### Problem as AI Task
[Classification / ranking / recommendation / generation /
forecasting / other]

### Data Sources
- Internal
- External
- User-generated

### Labeling / Ground Truth
[How will labels or truth signals be created?]

### Baseline
[What simple alternative will you compare against?]

### Model / API Choice
[Why this model / API / rules approach?]

### Evaluation Framework
| Metric Type | Metric | Why It Matters |
|---|---|---|
| Technical | | |
| Business | | |
| Trust / Guardrail | | |

### Launch Thresholds
[What would be good enough to move forward?]
```

7.6 Business, Economics, and Scaling View

Purpose

Show that the idea can survive commercially and operationally.

Must include

- monetization logic
- cost logic
- unit economics thinking
- scaling constraints
- operations implications

Template

```
## Business, Economics, and Scaling
```

```
### Monetization Model
```

```
[Subscription / usage-based / hybrid / enterprise / internal  
value case]
```

```
### Value Logic
```

```
[How does the product create value?]
```

```
### Cost Drivers
```

- Model / API cost
- Support cost
- Data / operations cost
- Review cost

```
### Unit Economics View
```

```
[How do you think about margin or value sustainability?]
```

```
### Scaling Risks
```

- Latency
 - Cost growth
 - Workflow complexity
 - Governance burden
-

7.7 Ethics, Governance, and Risk Note

Purpose

Show responsible product judgment.

Must include

- bias
- privacy
- hallucination or unsafe output risk
- misuse risk
- controls
- what should block launch

Template

```
## Ethics, Governance, and Risk Note

### Main Risks
- Risk 1
- Risk 2
- Risk 3

### Harm Scenarios
[Who could be harmed and how?]

### Controls
- Guardrails
- Review process
- Escalation path
- Logging / auditability

### Launch Blockers
[What would cause you to pause or reject launch?]
```

7.8 Prototype / MVP

Purpose

Show the core product interaction or decision loop.

Acceptable forms

- web application
- browser AI prototype
- local AI prototype
- no-code prototype
- Wizard-of-Oz MVP
- human-in-the-loop demonstrator

Strong prototype characteristics

- narrow but meaningful
- tied to the product story
- shows the AI role clearly
- shows fallback / review logic where needed

Weak prototype characteristics

- generic chatbot
- copied demo with no business framing
- pretty interface with no real decision logic

Prototype template

```
## Prototype / MVP
```

```
### Prototype Type
```

```
[Web app / no-code / browser AI / Wizard of Oz / other]
```

```
### What It Demonstrates
```

```
[What core workflow or decision loop is shown?]
```

```
### AI Role
```

```
[Where exactly does AI create value?]
```

```
### What It Does Not Yet Prove
```

```
[Be honest about the limitations]
```

```
### Links
```

- Repository:
- Live Demo:
- Screenshots / Video:

Important note on prototype ambition

Your prototype does **not** need to be fully production-grade.

A strong prototype may use:

- browser-based AI workflows
- API-based orchestration
- no-code or low-code tools
- mocked integrations
- staged automation with human review

What matters is whether the prototype helps us judge:

- the user experience,
 - the role of AI in the workflow,
 - the trust model,
 - and whether the product deserves pilot investment.
-

7.9 AI Collaboration Log

Purpose

Show that AI was used thoughtfully, not blindly.

Mandatory template

```
## AI Collaboration Log

### Tools Used
- ChatGPT / Claude / Gemini / Copilot / Cursor / v0 / Replit /
other

### Where AI Helped
- discovery
- drafting
- wireframing
- coding
- metrics
- risk review

### Example Prompts or Workflows
- Prompt 1:
- Prompt 2:
- Prompt 3:

### Where AI Was Weak or Wrong
- generic outputs
- hallucinated facts
- weak prioritization
- unrealistic recommendations

### Human Corrections
- what we changed
- why we changed it
- what judgment was applied

### Final Reflection
- where AI accelerated PM work
- where AI could not replace PM reasoning
```

8. How PM Artifacts Change for AI Products

Your submission must make this visible.

Standard PM Artifact	What Changes in an AI Product
Problem Statement	Must define the decision/workflow being improved
Persona / User Story	Must account for uncertainty, trust, and error cost
PRD	Must include data assumptions, fallback logic, and risk
Roadmap	Must include staged learning and proof gates
Metrics	Must connect model metrics to business outcomes
Launch Checklist	Must include review, rollback, and escalation
Pricing Note	Must account for inference or operating cost
Review Memo	Must separate hype from defensible product logic

Also show how AI helped create these artifacts:

- summarizing feedback
- drafting PRD structure
- generating prioritization alternatives
- producing wireframe starting points
- supporting prototype coding
- generating scenario analysis

But do not outsource these decisions to AI:

- opportunity choice
- risk acceptance
- launch recommendation
- monetization judgment
- governance claims

9. Cases You Can Use

Use at least one course case as a reference point.

Recommended case lenses:

- **Swiggy** - invisible AI, ETA, marketplace logic
- **Google Maps** - AI stack and product coordination
- **Zomato** - strategy and opportunity choice
- **Amazon vs ChatGPT** - feature vs platform logic
- **Google SGE / Yellow.ai** - UX, prompts, assistants, agents
- **iMerit** - data operations and labeling quality
- **Instacart / Tesla** - lifecycle and drift

- **Netflix** - metrics and evaluation
- **OpenAI Scaling** - cost, latency, scaling
- **Bias in Lending / Apple Card lens** - ethics and governance
- **OpenAI Pricing** - monetization and API economics
- **Microsoft / Nadella culture** - culture and cross-functional execution

Use the case to answer:

- what is similar?
 - what is different?
 - what lesson transfers?
 - what would your team do differently?
-

10. Demos You Can Use

Local demos in this workspace

- `Demos_Python\\Session01_Demo_ETA_Prediction.py`
- `Demos_Python\\Session02_Demo_AI_Stack_Simulation.py`
- `Demos_Python\\Session03_Demo_Opportunity_Matrix.py`
- `Demos_Python\\Session04_Demo_Retention_Flywheel.py`
- `Demos_Python\\Session05_Demo_Prompt_Engineering.py`
- `Demos_Python\\Sessions_09_12_Demos.py`

Shared demo repo

- [VinayaSharada/KateelLearningDemosToStudents](#)

Use demos:

- to understand a product pattern
- to prototype a PM decision support tool
- to test an AI workflow
- to generate evidence for a product choice

Do not submit a demo clone without product reasoning.

If you use a demo, explain:

- why you chose it
 - what PM question it helped you answer
 - what you changed
 - what it still cannot prove
-

11. Suggested Milestones

Topic Approval Form

Use this one-page form before you lock your project.

Topic Approval Form

Team Name

[Insert team name]

Team Members

- [Name 1]
- [Name 2]
- [Name 3]

Working Title

[Proposed product / feature / workflow title]

Industry / Function

[Finance / HR / retail / healthcare / operations / other]

Target User

[Who will use or benefit from this?]

Current Workflow Problem

[What is slow, costly, inconsistent, or frustrating today?]

Why This Matters

[Why is this a meaningful business problem?]

Proposed AI Role

[Prediction / recommendation / generation / classification / triage / copilot / agentic workflow]

Why AI Instead of a Simpler Alternative

[Why rules, search, analytics, or process redesign alone are not sufficient]

Human-in-the-Loop Design

[Where must people still review, approve, override, or escalate?]

Smallest Credible MVP

[What narrow wedge will you actually prototype?]

Build vs Buy vs API View

[Initial recommendation and why]

Main Risks

- [Risk 1]
- [Risk 2]
- [Risk 3]

Success Evidence

[What signals would tell you this idea is promising?]

Team Recommendation

[Proceed / refine / change topic / reject]

Milestone 1: Topic and Strategy Freeze

Submit:

- team members
- topic
- product vision
- JTBD / problem framing
- early build vs buy view

Milestone 2: Artifact Review

Submit:

- PRD draft
- wireframes
- MVP definition
- roadmap
- data/model assumptions

Milestone 3: Evaluation and Risk Review

Submit:

- metrics
- scaling / economics note
- ethics / governance / risk note

Final Submission

Submit the full package.

Presentation and pitch expectation

Your final presentation should feel like a serious product review, not a classroom recital.

It should clearly answer:

- what problem matters here?
- why this user and why now?
- why AI?
- why this MVP?
- why should the organization fund, pilot, or reject this idea?

Treat the faculty audience as if they are a leadership review group deciding whether your product moves forward.

12. Evaluation Rubric

Total suggested marks: **30**

Criterion	What Excellent Work Looks Like
Product framing	Sharp problem, narrow wedge, clear AI justification
Strategy	Strong opportunity logic and sensible build/buy decision
PM artifacts	Clear, useful, AI-aware PM documents
UX / workflow	Good handling of uncertainty, trust, and fallback
Data / model / evaluation	Realistic assumptions and meaningful metrics
Business / scaling	Clear value logic, cost logic, and operational realism
Ethics / governance	Specific risks, controls, and launch discipline
Prototype	Demonstrates the core value credibly
AI collaboration	Transparent, thoughtful, and critical use of AI tools

Common failure patterns

- vague problem
- hype-driven solution
- generic PM artifacts
- copied framework usage
- prototype disconnected from the strategy
- weak risk thinking

- raw AI output presented as team thinking

Final Presentation Scoring Sheet

Use this as a self-check before your final presentation.

Dimension	What the audience should feel or learn	Self-check question
Problem clarity	The team understands a real business problem	Would a leader immediately understand why this matters?
User realism	The user and workflow feel credible	Have we described an actual professional context, not a generic persona?
AI justification	AI is clearly the right lever	Have we explained why this is not better solved with rules, dashboards, or manual process changes?
Product judgment	Trade-offs are visible	Have we made hard choices about scope, sequencing, and controls?
Prototype credibility	The demo proves something important	Does the prototype show the core value, not just surface polish?
Metrics and evidence	Success is measurable	Can we explain what we would track in pilot and why?
Risk maturity	The team understands downside risk	Have we shown what could go wrong and how we would control it?
Executive recommendation	The close is decisive	Are we clearly saying pilot, iterate, scale, or stop?

Recommended presentation structure

For a strong final review, structure the story in this order:

1. problem and business stakes
2. target user and workflow
3. why AI, and why now
4. product concept and MVP scope
5. core PM artifacts and decisions
6. prototype walkthrough
7. metrics, economics, and launch risks
8. final recommendation

Avoid ending with a demo that is disconnected from the product argument.

13. Team Roles

Suggested roles:

- product strategy lead
- PM artifacts / PRD lead
- UX and workflow lead
- data / evaluation lead
- prototype / build lead
- business / risk lead

You must disclose:

- who led each artifact
 - who built what
 - how the team used AI
 - and how the team validated outputs together
-

14. Academic Integrity and Professional Standards

You may use AI tools extensively, but:

- do not submit raw AI output as your final work
- disclose meaningful AI help
- verify facts and assumptions
- distinguish evidence from speculation
- do not fabricate research, data, or product behavior

Professional-quality work means:

- clear reasoning
 - explicit trade-offs
 - honest assumptions
 - responsible recommendations
-

15. Final Submission Checklist

Before submitting, check:

- ☐ The problem is specific
- ☐ The AI role is clear

- ☐ The product strategy is defensible
 - ☐ The PM artifacts are complete
 - ☐ The UX handles uncertainty
 - ☐ The evaluation logic is meaningful
 - ☐ The business and scaling note is realistic
 - ☐ The ethics/risk note is specific
 - ☐ The prototype shows the core value
 - ☐ The AI collaboration log is honest
 - ☐ The recommendation is clear: stop / iterate / pilot / scale
-

16. What Excellent Teams Sound Like

Excellent teams sound like this:

- “We chose this workflow because it has repeated friction, measurable value, and a realistic MVP path.”
- “We used AI to speed up drafting and prototyping, but not to outsource judgment.”
- “We recommend API-first for MVP because learning speed matters more than full model ownership right now.”
- “The prototype is intentionally narrow because the real risk is trust and workflow fit.”
- “We would not scale yet because evaluation and launch controls are not strong enough.”

That is the standard to aim for.

17. Final Advice

- Start with the workflow problem.
- Keep the first version narrow.
- Use frameworks only when they improve a decision.
- Use AI as an accelerator, not as a substitute for thinking.
- Make your PM artifacts usable by a real team.
- Treat the prototype as evidence, not theater.

The best submissions will show that your team knows how to **manage an AI product**, not just describe one.

18. Good Topics vs Weak Topics

These examples are meant to sharpen judgment, not limit creativity.

Example 1: Finance Operations

Weak version:

"An AI chatbot for finance teams"

Why it is weak:

- unclear workflow
- unclear user
- unclear business value
- no explanation of why chat is the right interface

Better version:

"An AI assistant for accounts payable teams that extracts invoice data, flags mismatches, recommends next actions, and routes low-confidence cases to human reviewers"

Why it is stronger:

- clear workflow
- repeated decision pattern
- visible human-in-the-loop role
- measurable value through cycle time, exception handling, and error reduction

Example 2: Product Management Work

Weak version:

"An AI tool for product managers"

Why it is weak:

- too broad
- sounds like a category, not a product wedge
- impossible to evaluate in one capstone

Better version:

"A feature prioritization copilot that helps B2B PMs compare backlog items using RICE, customer pain signals, support evidence, and implementation effort assumptions"

Why it is stronger:

- narrow decision use case
- direct link to PM work
- easy to prototype
- clear room for judgment rather than blind automation

Example 3: Retail or Commerce

Weak version:

"A recommendation engine for e-commerce"

Why it is weak:

- too generic
- no clear business context
- no wedge, metrics, or operational assumptions

Better version:

"A post-purchase retention recommender for an online beauty retailer that identifies at-risk customers and recommends the next best retention intervention for the CRM team"

Why it is stronger:

- defined moment in the customer lifecycle
- clear user of the output
- business outcome is easier to measure
- more realistic MVP than a full-site personalization engine

Rule to remember

If the topic can be explained in one vague sentence, it is probably still too broad.

If the topic makes a concrete workflow, user, decision, and value path visible, it is much more likely to become a strong project.

Part B. Sample Strong Submission Outline

19. How to Use This Sample

This sample is **not** a model answer to copy.

Use it to understand:

- the expected level of specificity,

- the tone of a strong submission,
 - and how the different parts of the assignment connect.
-

20. Hypothetical Project Example

Title

InvoiceIQ: An AI-Powered Accounts Payable Triage and Exception Resolution Assistant

One-line concept

An AI-enabled workflow assistant for enterprise finance teams that:

- ingests vendor invoices,
- extracts and classifies key fields,
- flags mismatches and risk conditions,
- recommends actions,
- and routes low-confidence or high-risk cases for human review.

Why this is a good sample topic

It has:

- a real business workflow,
 - measurable operational pain,
 - clear value,
 - repeated decisions,
 - realistic AI use,
 - trust and governance implications,
 - and a sensible MVP path.
-

21. Sample Strong Submission Structure

21.1 Executive Summary

Problem

Large enterprises process thousands of invoices manually. Teams spend time on:

- extracting fields from semi-structured invoices,

- matching them against purchase orders and goods receipts,
- identifying duplicates or suspicious changes,
- and escalating exceptions to the right reviewer.

This creates:

- slow processing,
- delayed payments,
- vendor dissatisfaction,
- fraud exposure,
- and high finance operations cost.

Proposed Product

InvoicIQ is an AI-powered accounts payable assistant that:

- extracts invoice data,
- checks it against enterprise records,
- flags anomalies,
- recommends a next action,
- and routes cases by confidence and risk.

Why AI?

Traditional rule-based automation works for clean, repetitive cases but struggles when:

- invoice formats vary,
- vendors change layouts,
- exceptions require interpretation,
- and duplicate/fraud patterns are subtle.

Recommendation

Proceed with a **narrow pilot MVP** focused on:

- one business unit,
- one invoice category,
- and one set of exception rules.

Do not scale broadly until:

- extraction quality,
- review workflow fit,
- and escalation accuracy

are proven.

21.2 Product Strategy

Vision

Reduce accounts payable cycle time and exception-handling effort by giving finance teams a reliable AI-assisted review workflow rather than a fully autonomous black box.

JTBD

When I am processing a large volume of invoices with varying formats and possible mismatches,
I want to identify which invoices can move forward safely and which ones need review,
so that I can reduce manual effort, avoid payment errors, and maintain control over financial risk.

Target Users

- AP operations analyst
- AP team lead
- finance controller
- procurement reviewer

Opportunity Matrix

AI Feature	Business Value	Feasibility	Priority
Invoice field extraction	High	High	High
PO / GRN mismatch detection	High	Medium	High
Duplicate invoice risk scoring	High	Medium	High
Vendor query generation	Medium	High	Medium
Cash-flow forecasting	Medium	Low	Low for MVP

Build vs Buy vs API

Recommendation: API-first plus workflow orchestration

Why:

- extraction APIs already exist,
- custom training is not needed for MVP,
- differentiation lies in workflow integration, confidence routing, and exception logic.

First Wedge

Start with:

- domestic vendor invoices,
 - non-complex procurement categories,
 - one business unit
-

21.3 PM Artifact Pack

Problem Statement

Accounts payable teams waste time manually reviewing invoices that could be auto-triaged, while still missing subtle exceptions that matter.

Core workflow

1. invoice arrives
2. fields extracted
3. invoice checked against records
4. confidence and anomaly score generated
5. action recommendation shown
6. invoice is approved, reviewed, or escalated

Prioritization Table

Feature	User Value	Effort	Confidence	Priority
Invoice extraction	High	Medium	High	High
Confidence-based routing	High	Medium	High	High
Duplicate detection	High	Medium	Medium	High
Exception summary	Medium	Low	High	Medium
Vendor follow-up drafting	Medium	Low	High	Medium

MVP definition

The MVP is successful if it can:

- extract fields from a narrow document set,
- route low-confidence cases correctly,
- reduce review time,
- create trust among finance reviewers

Roadmap

- Phase 1: workflow discovery
 - Phase 2: MVP
 - Phase 3: controlled pilot
 - Phase 4: scale
-

21.4 UX / Workflow / Trust Design

UX principle

The system should feel like a **review assistant**, not an overconfident autonomous controller.

Key screens

- invoice list with triage status
- invoice detail view
- extracted fields with confidence
- anomaly panel
- recommended next action
- human override and reason capture

Confidence handling

- High confidence: fast approval support
- Medium confidence: recommendation plus uncertainty
- Low confidence: route to manual review

Human-in-the-loop

The reviewer must be able to:

- correct extracted fields

- override recommendation
 - record a reason
-

21.5 Data, Model, and Evaluation Strategy

AI task decomposition

This is a workflow using multiple components:

- OCR/document extraction
- entity extraction
- rule-based validation
- anomaly scoring
- exception classification
- recommendation generation

Data sources

- historical invoices
- purchase order records
- goods receipt records
- vendor master data
- exception resolution logs

Metrics

Technical metrics

- field extraction accuracy
- exception classification precision / recall
- duplicate detection precision

Business metrics

- reduction in review time
- straight-through processing rate
- reduction in payment delay
- reduction in duplicate or erroneous payments

Trust / guardrail metrics

- override rate
- false auto-approval rate
- reviewer complaint rate

21.6 Business, Economics, and Scaling

Value logic

The product creates value through:

- lower manual review effort
- faster turnaround
- better exception detection
- reduced duplicate/fraud risk

Cost drivers

- document processing cost
- model/API cost
- integration and maintenance
- reviewer time for exception cases

Unit economics logic

The product works economically only if:

- low-risk cases move faster,
- reviewer time drops meaningfully,
- AI cost per invoice stays below savings created

21.7 Ethics, Governance, and Risk

Key risks

- false approval of incorrect invoice
- duplicate payment not caught
- weak explainability causing blind trust
- privacy/confidentiality issues

Controls

- no full autonomy for risky categories
- confidence thresholds
- mandatory review for certain exceptions
- audit logs
- override capture

Launch blocker

If false auto-approval risk cannot be kept acceptably low, the system should not move beyond a narrow pilot.

21.8 Prototype / MVP

Example MVP scope

A web-based prototype where a reviewer can:

- upload a sample invoice
- see extracted fields
- see field confidence
- see mismatch summary
- receive a recommendation
- choose approve / review / escalate

What it proves

- workflow clarity
- AI role in triage
- usefulness of confidence routing
- trust and override behavior

What it does not prove yet

- enterprise-scale economics
 - full ERP integration
 - production-grade extraction quality across all formats
-

21.9 AI Collaboration Log

Example strong usage

Tools used

- ChatGPT for framing and PRD drafting
- Copilot / Cursor for prototype coding
- AI UI tool for wireframe starting points

Good use

- first-pass JTBD options
- summary of workflow interviews
- draft UI text
- edge-case test scenarios

Where AI was weak

- generic personas
- unrealistic automation assumptions
- weak governance framing
- poor data-quality realism

Human correction

- narrowed the workflow
 - added reviewer override logic
 - changed full automation to assisted triage
 - made the launch recommendation more conservative
-

22. Why This Would Be Strong

This is strong because:

- the workflow is real and valuable
 - the AI role is specific
 - the MVP is narrow and credible
 - the PM artifacts are real
 - the metrics connect to operations and business value
 - the risk note is serious
 - the prototype proves something meaningful
 - the team uses AI but does not blindly trust it
-

23. What Would Make It Weak

The same project would become weak if the team:

- turned it into a generic chatbot
- skipped the data assumptions
- ignored reviewer trust
- claimed "90% automation" with no evidence

- produced a nice UI but no launch logic
 - used AI-generated content without critique
-

24. Final Message

Use the sample as a reference for:

- specificity
- structure
- product judgment
- realism

Do not copy its topic or wording mechanically.

The strongest submissions will show that your team understands how to:

- define the right problem,
- choose the right AI role,
- create strong PM artifacts,
- build a meaningful MVP,
- and make a responsible product recommendation.